CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE-201 CONTROL ENGINEERING

Date: 29.11.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **(a)** Define open loop and closed loop system by giving suitable examples **[05]**

(b) Define the transfer function of a system. Write advantages of transfer function. **[02]**

Q - 2 **(a)** Obtain the transfer function C/R for the block diagram shown in fig.1. **[06]**

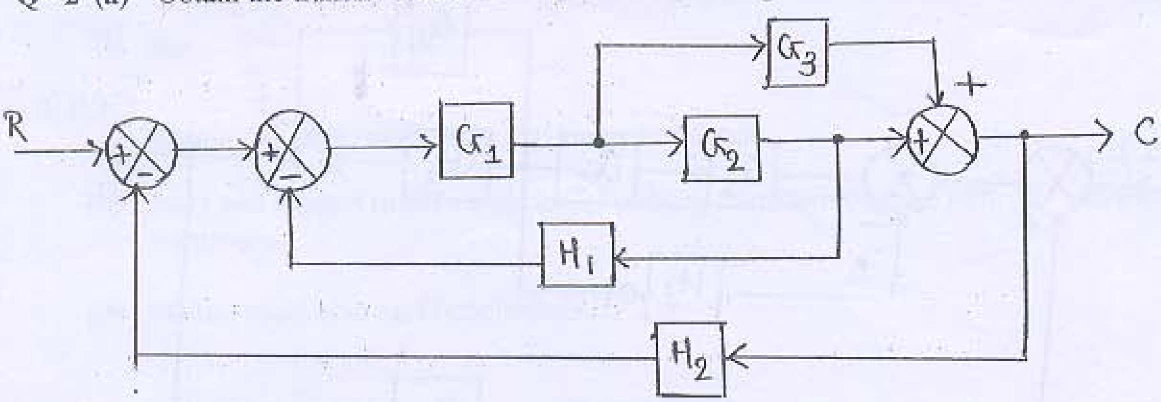


Fig. 1.

(b) Find the transfer function of the system shown in fig.2 using Mason's gain formula. **[06]**

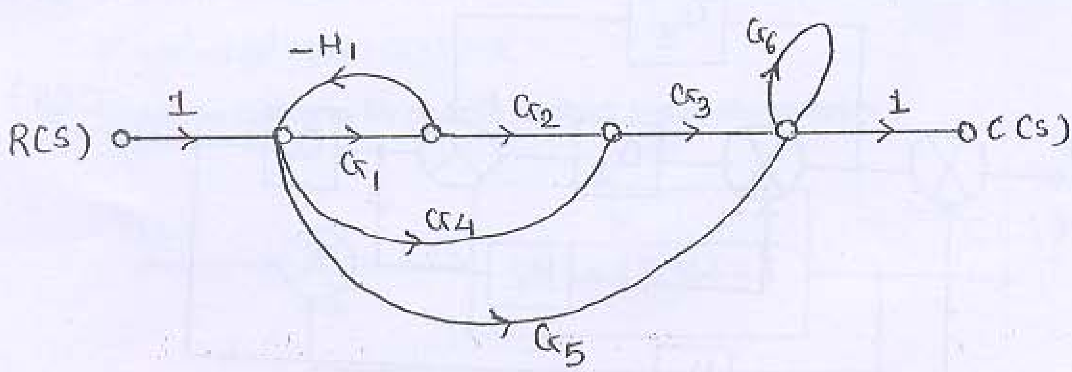


Fig. 2.

(c) The signal-flow graph shown in fig.3. Obtain total number of forward paths and total number of non-touching loops. [02]

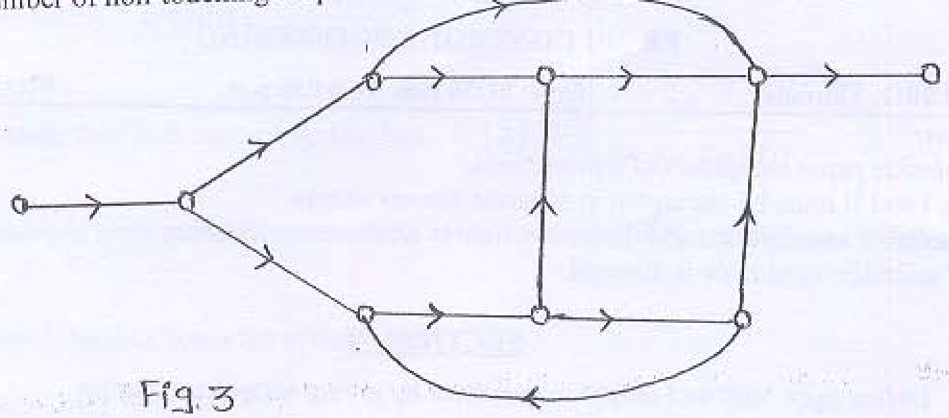


Fig. 3

OR

Q - 2 (a) Draw the corresponding signal flow graph of given fig.4 and find C(S)/R(S). [06]

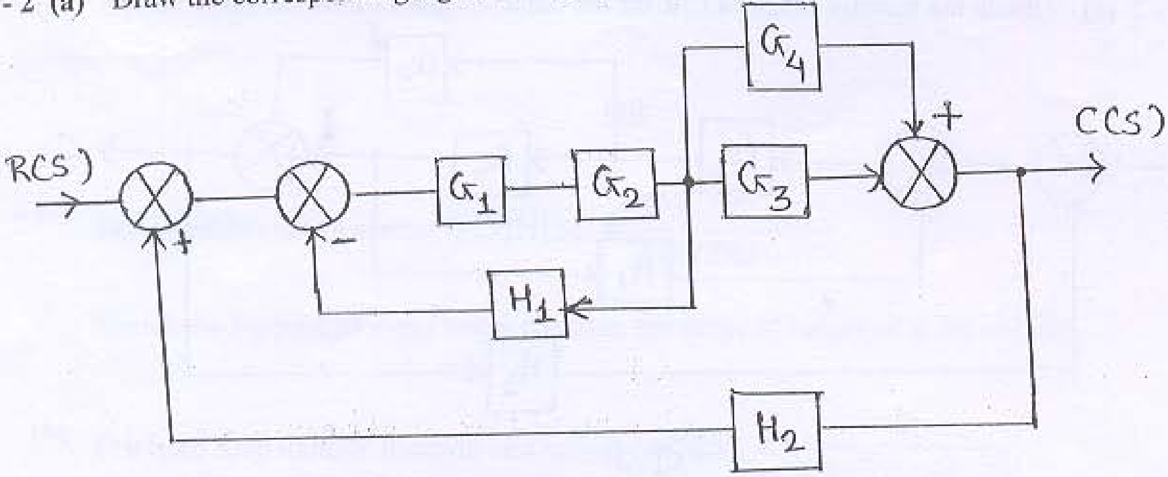


Fig. 4.

(b) Reduce the block diagram shown in fig.5 and obtain its transfer function. [06]

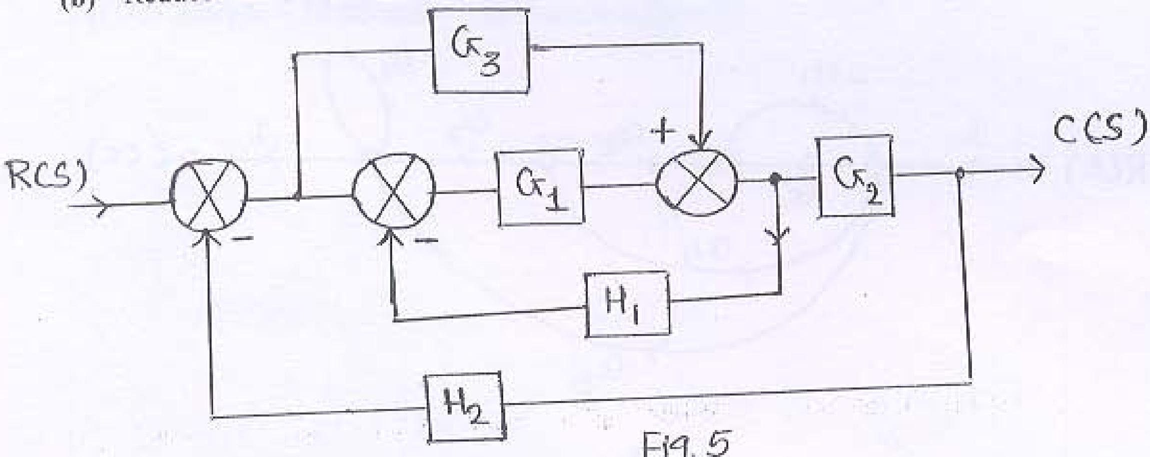


Fig. 5

(c) Give any two difference between block diagram and signal flow graph methods. [02]

Q - 3 (a) Find the controllability of the system described by the state equation [05]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

(b) The control system having feedback has [05]

$$G(S) = \frac{20}{S(1+4S)(1+S)}$$

Determine: (i) Different static error coefficients

(ii) Steady state error if input = $r(t) = 2+4t + t^2/2$

(c) Explain standard test signals. [04]

OR

Q - 3 (a) For, $\frac{G(S)}{R(S)} = \frac{1}{S^2+S+1}$ [05]

Obtain rise time, peak time, maximum overshoot.

(b) State and explain routh's criterion of stability also state the two difficulty and their solution. [05]

(c) Write a short note on PI controller. [04]

SECTION- II

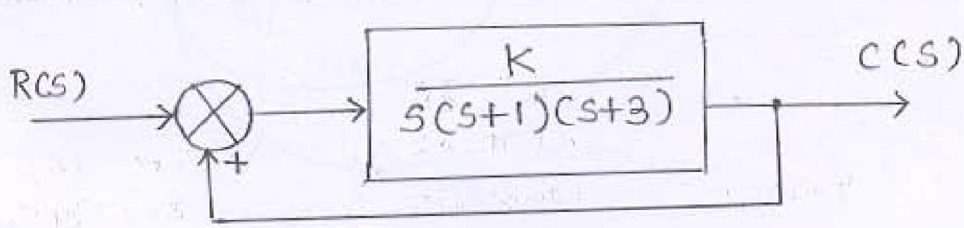
Q - 4 (a) Explain effect of change of G(s)H(s) on steady state error. [05]

(b) $S^3 + 4S^2 + S + 6 = 0$, state how many roots are in right half of S-plane. [02]

Q - 5 (a) Determine the range of K for stability of a system having characteristic equation [07]

$S^4 + 6S^3 + 12S^2 + 10S + (K+3) = 0.$

(b) Sketch the root locus for positive feedback system shown below : [07]



OR

Q - 5 (a) By applying Nyquist stability criterion, find the value of K for which the system is just stable. $G(S)H(S) = \frac{K(S+2)}{(S+1)(S-1)}$ [07]

(b) A unity feedback control system has $G(S) = \frac{80}{S(S+2)(S+20)}$ [07]

Draw the bode plot. Determine G.M., P.M. comment on the stability.

Q - 6 (a) Sketch the root locus for system with [07]

$$G(S)H(S) = \frac{K(S+4)}{S(S^2+2S+2)}$$

(b) Find the response of first order system . [07]

OR

Q - 6 (a) For a certain control system $G(S)H(S) = \frac{K}{S(S+2)(S+10)}$ [07]

Sketch the Nyquist plot and hence calculate the range of values of K for stability.

(b) The open loop transfer function of a system is, $G(S) = \frac{K}{S(1+S)(1+0.1S)}$ [07]

Determine the values of K such that

i) Gain margin = 10 dB and ii) Phase margin = 50°

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